EXPERIMENT NOTES

Garrett McNeill

INITIAL PARTITIONING

- LegalBench is the first benchmark task I will evaluate Gemma2B on to gain a baseline
- LegalBench has around ~162 task to evaluate on
- These are partitioned into 6 separate categories

- I want to train a model on each subtasks
- This will result in 6 CRLs
- Need to save parameter networks for each CRL
- Once we finish training, evaluate on benchmark the model was trained on
- Then cross evaluate on the other legal bench benchmarks

CODE EXECUTION FOR LEGALBENCH

- First try testing on one subset, then expand to others
- Begin with all subsets / tasks that include binary classification
- CRL process is going to need a quick way to evaluate performance
- Limits us with more qualitative answering

SUBCATEGORIES

- Issue Spotting 17 (all yes/no)
- Rule Recall 5 (2 yes / no)’
- Rule Application and Rule Conclusion 12 (9 yes / no , 1 binary)
- Interpretation
- Rhetorical understanding

TWO MODELS

- Base model and fine tuned model (gemma2b-it)
- Can’t use fine tuned model for CRL since weights are different even though structure is the same
- Issue since fine tuned model produces more “interpretable” answers
- Does give room for interesting question - can CRL close the gap between fine tuning and how do the performances compare?

EVALUATION

- Non fine-tuned model produces ‘garbage’ answers sometimes
- Can measure how CRL limits such ‘garbage’ answers
- Since task are binary classification (for now) we can use confusion matrix for evaluation of models